



GE Healthcare

Technical Publication

Direction 5133397-100

Revision 3

GDAS POWER SUPPLY UPGRADE KIT

OPERATORS DOCUMENTATION

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REVISION HISTORY

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Revision	Date	Reason for change
1	04/22/05	Initial Release
2	05/05/05	Removed duplicate information in BOM Parts list.
3	06/15/2006	Added a note for reusing the existing 5V cable for 4 and 8 slice systems.

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SECTION 1 - INSTALLATION OVERVIEW

1.1 - OVERVIEW

This procedure updates a gantry containing GEMS-IT (Also referred to as GDAS-GE) GDAS Power Supplies to the new Astec GDAS power supply configuration. If either GEMS-IT power supply fail, this upgrade must be performed. A mixture of two different supplies, (Astec-GE and GEMS-IT) in the same gantry is not a supported configuration.

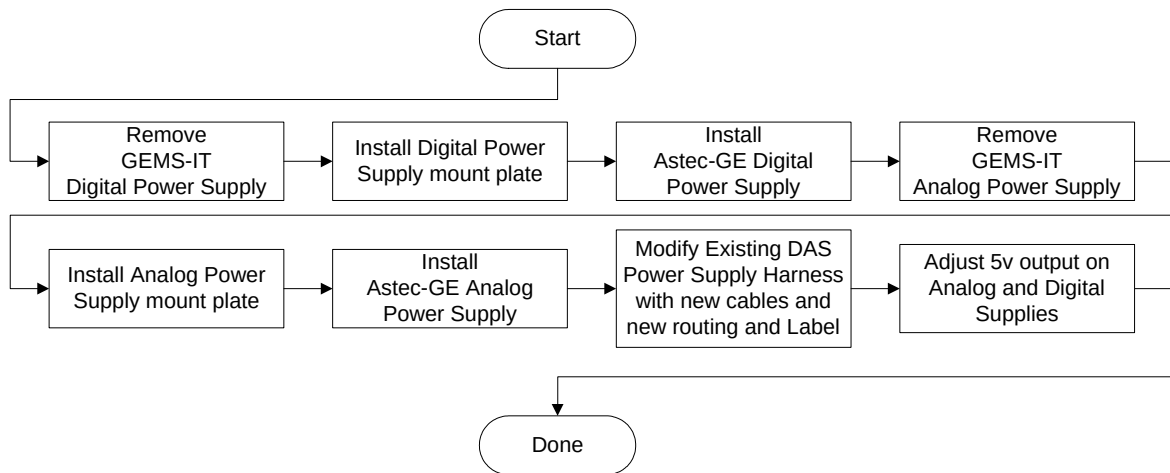


Illustration 1: Process Flowchart

1.1.1 VISUAL PERSPECTIVE OF THE FOLLOWING PROCEDURE

All left/right orientation in this procedure is based on the viewpoint of an individual looking through the gantry opening, from the patient table end of the system.

The digital and Analog GDAS power supplies are mounted to the rotating base of the gantry. Looking through the gantry from the table end, the power supplies are more easily serviced from the left side of the gantry.

1.1.2 REQUIRED TORQUE SPECIFICATION FOR #4-40 CONNECTOR JACKSCREWS. (BACKPLANE AND POWER SUPPLIES)

All cable connector jackscrews should be tightened according to the following specifications:

Table A: Cable Jackscrew Torque specification	
#4-40 Jackscrew Specification	
N-m	Inch-lbs
0.45	4.0

1.2 – PERSONNEL REQUIREMENTS

Required Persons: 1 (See Caution Statement)
 Procedure execution time: 110 minutes

Finalization time: 10 minutes

1.3 – REQUIRED TOOLS

Table 1: Required Tools

Item	Qty	Effectivity	Part#	Manufacturer
Torque Wrench	1	-	-	-
Torque Screwdriver	1	-	-	-
Standard FE Toolkit	1	-	-	-
Short (Stubby) Medium flat blade screwdriver	1	-	-	-
Short Small flat blade Screwdriver	1	-	-	-
Medium Phillips (+) head Screwdriver	1	-	-	-
Regular medium sized Slip Joint Pliers	1	-	-	-
Medium size Side Cutters	1	-	-	-
3/8 inch Ratchet Drive wrench set	1	-	-	-
12 inch (30cm), 3/8 inch extension or equivalent adding up to 12 inches (30cm)	1	-	-	-
Torx Wrench set (T15, T30, T25)	1	-	-	-
3/8 inch drive, Metric Hex Wrench set (12mm and 10mm Hex)	1	-	-	-
3/8 inch drive, Metric Hex Wrench set (12mm and 10mm Hex)	1	-	-	-
Insulated "Tweaker" Adjustment Tool (6-inch length)	1	-	-	-

1.4 – CONSUMABLES

- These should be obtained locally.

Table 2: Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Loctite 271 (Red) or equivalent	50cc	-	-	-
Loctite 242 (Blue) or equivalent	50cc	-	-	-

1.5 – REPLACEMENT PARTS

- These are the FRU part numbers. Upon failure order these specific replacements.

Table 3: Replacement FRU parts

Item	Qty	Effectivity	Part#	Manufacturer
Astec Digital Power Supply	1	All GDAS	5126615	Astec
Astec Analog Power Supply	1	All GDAS	5126617	Astec

SECTION 2 – SAFETY

⚠ DANGER

ELECTROCUTION HAZARD.

HIGH VOLTAGE PRESENT.

USE PROPER LOCKOUT / TAGOUT PROCEDURES BEFORE REPLACING THE COMPONENTS.

⚠ DANGER

CRUSH HAZARD.

UNEXPECTED GANTRY ROTATION CAN CAUSE SERIOUS INJURY.

NEVER SERVICE THE GANTRY WITH THE AXIAL DRIVE ENABLED.

⚠ CAUTION

Potential for injury to service personnel.

Each power supply chassis is just under 30 pounds.

Depending on your height, a step stool may be required to reach over the TGP for supply removal/installation. Use your judgment to determine if a second person may be required to assist in removal and installation.

⚠ CAUTION

Burn injury possible.

Power supply may be hot to touch.

Use caution to prevent possible burn injury from hot surface contact.

NOTICE

Follow ALL required safety and PPE procedures customary for your organization when working on this product.

NOTICE

The Rotating Base within the Gantry must be configured with the same type of power supplies. Either two older types or two of the newer (Smaller) types must be used. **DO NOT** install two different types of power supplies on the same Rotating Base within the gantry.

SECTION 3 – KIT INVENTORY

This kit includes all the necessary hardware to upgrade the gantry to accommodate the use of new GDAS-Astec power supplies.

3.1 – KIT CONTENTS

1. Open the kit and inspect the contents for completeness and possible shipment damage.
2. To properly prepare for the upgrade, read this COMPLETE procedure at this time. Take the time to familiarize yourself with the upgrade. Especially section 5 DAS Harness Modification.

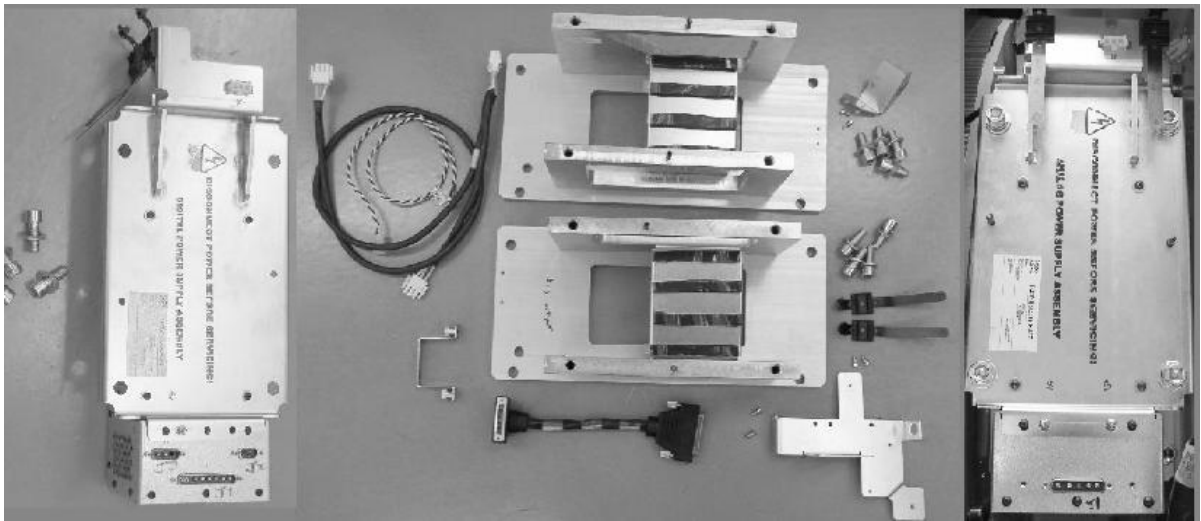


Illustration 2: GDAS Power Supply Upgrade Kit Contents (P/N 5133765)

3.2 – KIT INVENTORY CHECKLIST

Table 4: Inventory Checklist Kit P/N 5133765

#	✓	P/N	U/M	QTY	Description
1		5126615	EA	1.0	ASTEC Digital PS Assembly
2		5128120	EA	1.0	ASTEC Digital PS Mounting Asm.
3		5126617	EA	1.0	ASTEC Analog PS Assembly
4		5128121	EA	1.0	ASTEC Analog PS Mounting Asm.
5		2332018-4	EA	1.0	BULKHEAD, DAS FAN HTR, GDAS
6		2293442-4	EA	1.0	BRACKET, DET. HTR. HARNESS, ASTEC P.S. GDAS
7		5124434	EA	1.0	Cable, ASTEC Analog 5V To DAS
8		5124433	EA	1.0	Astec Power Supply AC Input Cable
9		5133770	EA	1.0	Jumper Cable, 24VDC ORP Power for Astec PS upgrade kit
10		5130042	EA	1.0	Bracket, Analog PS Conn. Restraint

11		46-325136P1	EA	2.0	CABLE CLAMP,RICHCO #UMS-16-45,.37 TO 1IN DIA.,RELEASABLE,MOUNTS WITH #8
12		46-328417P66	EA	8.0	SCREW HEXAGON SOCKET 12 MM 40 MM
13		46-328430P8	EA	8.0	WASHER PLAIN - NORMAL 13 MM 24 MM
14		46-328420P1	EA	2.0	SCREW BUTTON HEAD 4 MM 8 MM BLACK OXIDE, STEEL
15		46-328417P3	EA	4.0	SCREW HEXAGON SOCKET 4 MM 10 MM ZINC PLATED, STEEL
16		5134039	EA	1.0	Label- J3 Digital 24VDC Output
17		46-208758P3	EA	20.0	7.31~ X .184~ SELF-LOCKING CABLE TIE 1/16~ TO 1-3/4~ WIRE BUNDLE RANGE. 50LB TENSILE STRENGTH, NYLON MATR~L.
18		5133397-100	EA	1.0	GDAS Power Supply Upgrade Kit - Instructions
19		46-208758P10	EA	10.0	SELF-LOCKING CABLE TIE

Table 1: Kit inventory checklist

SECTION 4 – GEMS-IT DIGITAL POWER SUPPLY REMOVAL

The existing power supplies are manufactured by GEMS-IT. This name refers to the original Digital and Analog power supplies that shipped with systems that used the GDAS subsystem.

4.1 – GANTRY/TABLE PREPARATION

1. Position the table so it is out of the way.
2. Remove gantry side covers. Refer to Parts Replacement/Gantry/Enclosure/ (Cover Removal Procedure).



⚠ DANGER

ELECTROCUTION HAZARD.

HIGH VOLTAGE PRESENT.

USE PROPER LOCKOUT / TAGOUT PROCEDURES BEFORE REPLACING THE COMPONENTS.

3. From the right side of the gantry, turn off the Axial Drive, HVDC, and 120VAC Switches.
4. Remove the top covers.
5. Unplug the top cover fans.
6. Remove both top covers.
7. Rotate the gantry to access the GDAS-GE digital power supply from the left side. (Same side as TGP). See Illustration 3.

Position power supply in gantry to expose these cables and connectors

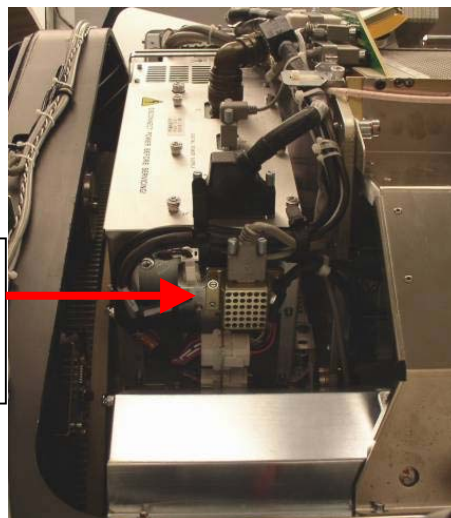


Illustration 3: Position Power Supply For Removal

8. Use the Index Lock to lock the gantry during this procedure. See Illustration 4.
9. Remove EMC cover from left side of Das chassis.

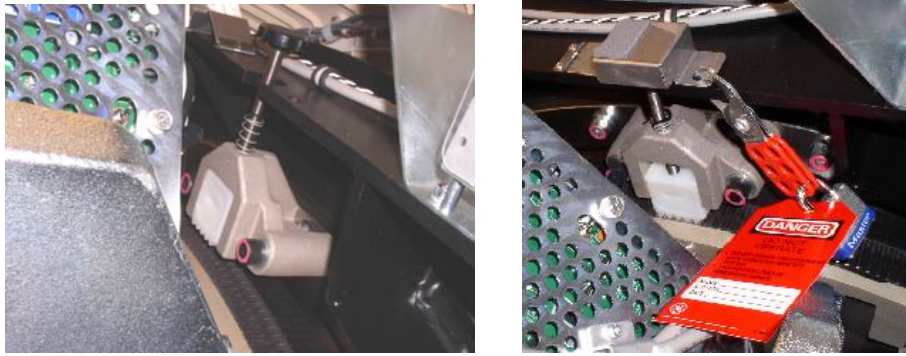


Illustration 4: (Safety) Index lock

10. Proceed to sub-section 4.1 GEMS-IT Digital Power Supply Removal.

4.2 – GEMS-IT DIGITAL POWER SUPPLY REMOVAL

Note: The cables are secured with re-useable cable clamps. DO NOT CUT THEM. Open the cable clamps by pulling back the small tab and pushing the strap back through the clamp. See Illustration 5.

1. Use the following illustration to remove the cables from the power supply.

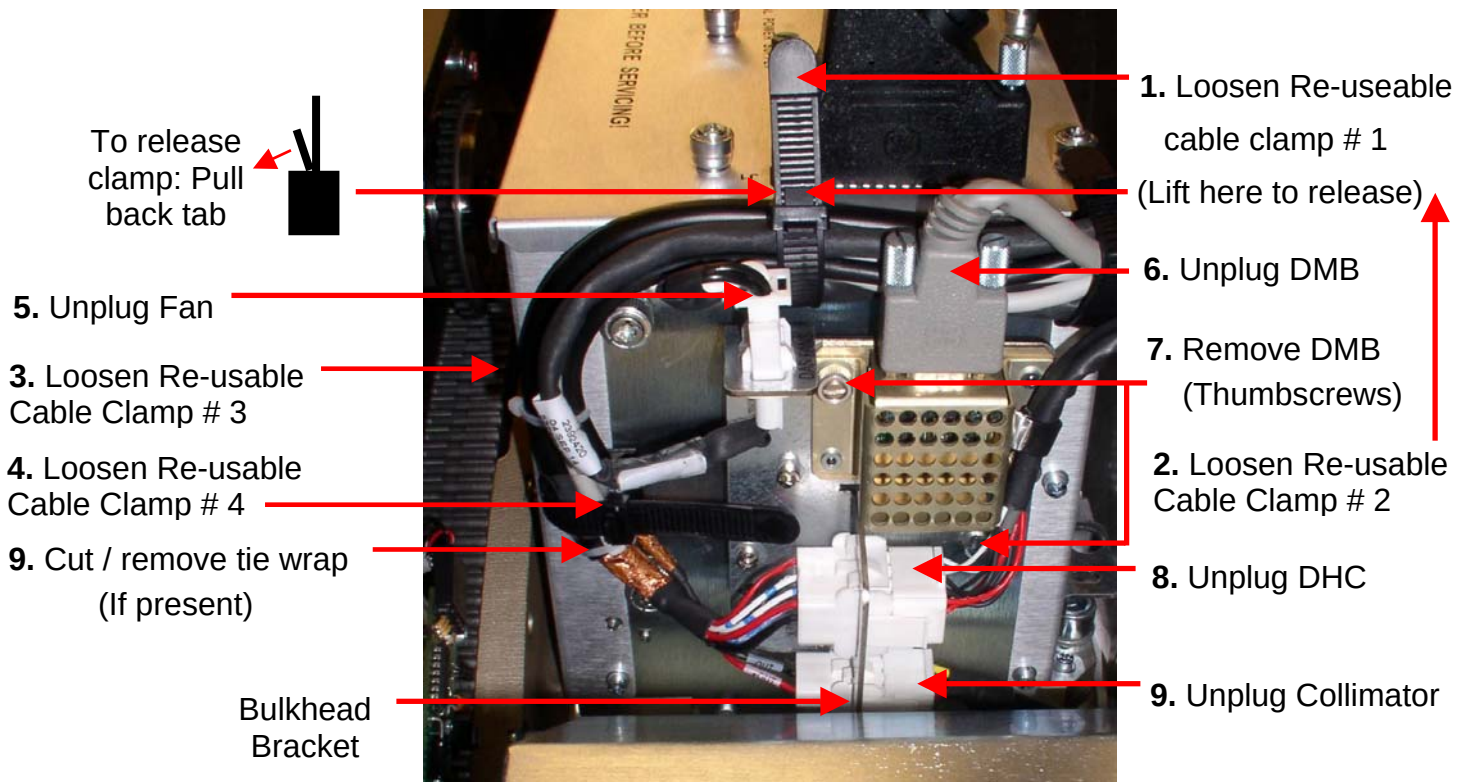


Illustration 5: Order to remove Hardware

2. The bulkhead bracket (connector mount) is part of the GEMS-IT power supply assembly. Because of this, it will be necessary to completely remove the Mate-N-Lok connectors from the bulkhead mount bracket using a small screwdriver. Follow the steps in illustration 6.

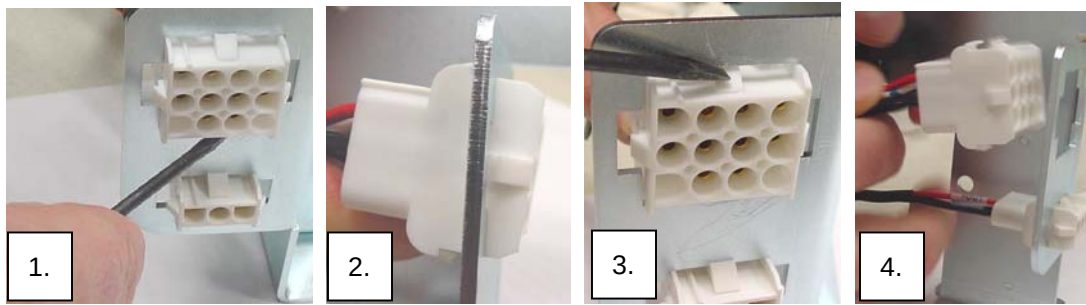


Illustration 6: Mate-N-Lok Connector Removal Order

3. Remove the Bottom Collimator connector using the same process as shown in Illustration 6.
4. Remove plugs J1, J2, J3 and J4 (120VAC) on the power supply.
5. Use a Phillips screwdriver to remove Cable Harness Bracket, (Captive screws). See illustration 7. (If necessary, temporarily remove the Index Lock and rotate the gantry).



Illustration 7: Cable Harness Bracket

6. Disconnect plugs J32, J31, J30, J49, and J33 of the GDAS chassis, in this order.
7. Completely remove the Cable Harness Bracket.
8. Use a Phillips Screwdriver. Remove the backplane connector restraint bracket. See illustration 8.



Illustration 8: Backplane Connector Restraint Bracket

9. Remove cables J99 and J104 from the Center GDAS chassis.
10. Disconnect J2 on Detector Heater Control Board (DHCB).
11. Disconnect J3 on DHCB.
12. Loosen the two black re-usable cable clamps on the main cable harness.
13. Disconnect J1 of the DHCB.
14. Move the harness out of the way. Slack in the harness can be achieved by cutting tie wraps. holding connector J32. Pull the harness up and over the left GDAS backplane. See illustration 9.

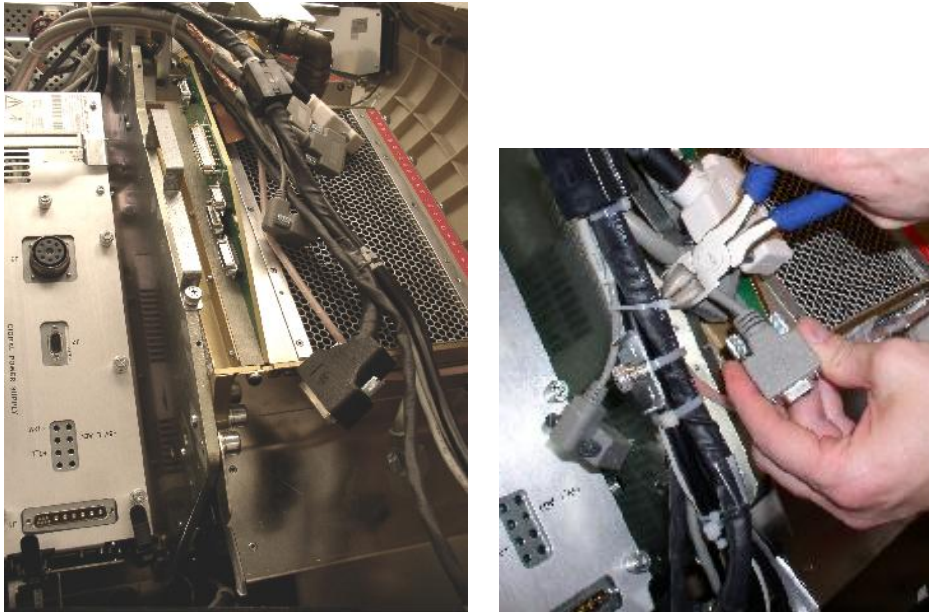
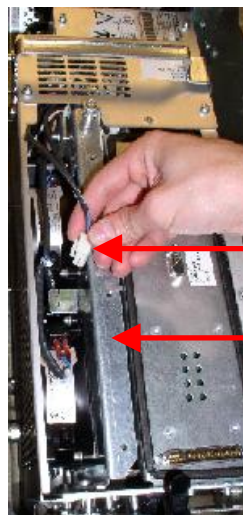


Illustration 9: DAS Harness moved out of the way- Necessary to cut tie wraps

15. Remove the top cover of the power supply, (7 captive screws) using a T25 Torx or medium flat blade screwdriver.
16. Loosen the 3 captive screws holding the fan tray assembly (1 on top and 2 on the side), using a 4mm Hex wrench, T30 Torx wrench or short flat blade screwdriver.
17. Disconnect the Fan tray power plug.
18. Slide the fan tray out of the power supply. See illustration 10.



1. Disconnect Power Plug.
2. Slide Fan Tray out of chassis

Illustration 10: Disconnect the Fan Tray Power Plug and Remove the Fan Tray

19. Using a 10mm Allen Hex bit and necessary extensions to completely loosen the four mounting bolts. See illustration 11.

Note: 4 captive mounting bolts and 1 guide pin holds the power supply to the rotating base. The guide pins will hold the power supply in position when all 4 bolts are removed.

20. Using a 10mm Allen Hex bit and necessary extensions to completely loosen the four mounting bolts. See illustration 11

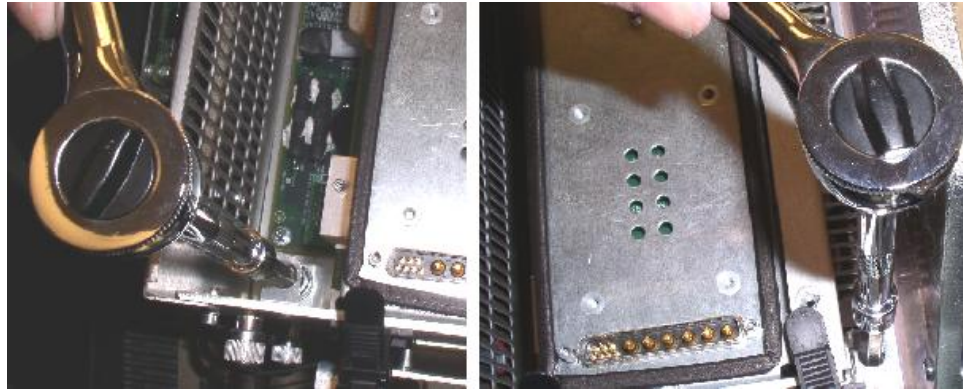


Illustration 11: Remove the 4 Power Supply Bolts

Note: Be careful not to hit the home flag sensor board when removing supplies on the TGP side of the gantry.

21. Remove the power supply assembly from the gantry. See Illustration 12.

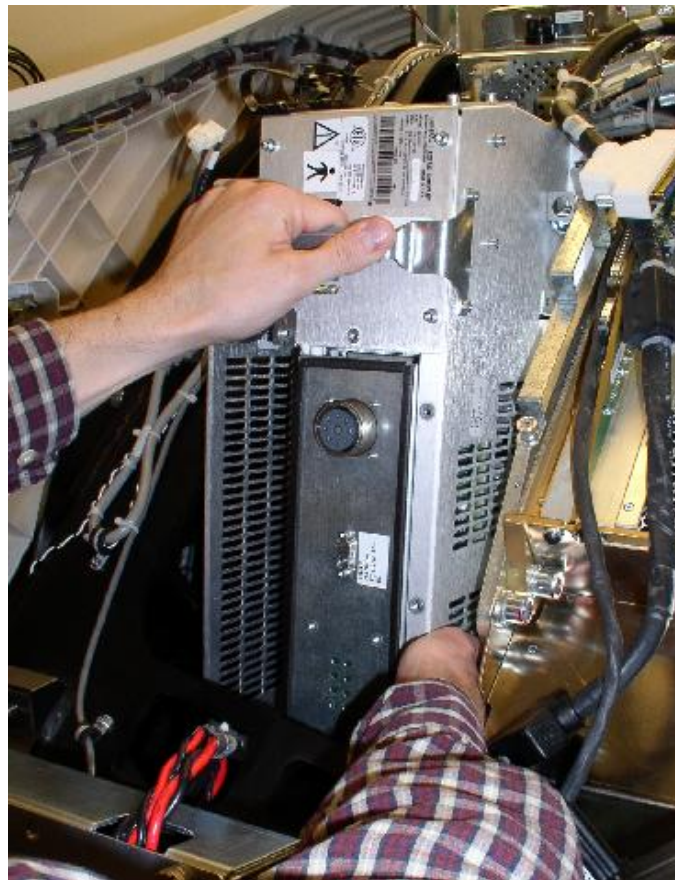


Illustration 12: Best leverage position of hands for Power Supply Removal

22. Proceed to Section 5 Astec Digital power Supply Installation.

4.3 –DIGITAL POWER SUPPLY MOUNT INSTALL

1. Install the re-useable cable clamps to the new DAS Fan/Htr/DMB bulkhead bracket Using black screws 46-328420p1. See illustration 13.



Illustration 13: GDAS Fan/Heater/DMP Bulkhead Bracket

2. Apply Loctite 242 to the two M4x10 mounting screws and install the bracket to the digital power supply mount base, (5128120). See illustration 14.
3. Torque the screws to 2.3 N-m with a 3mm Hex wrench.

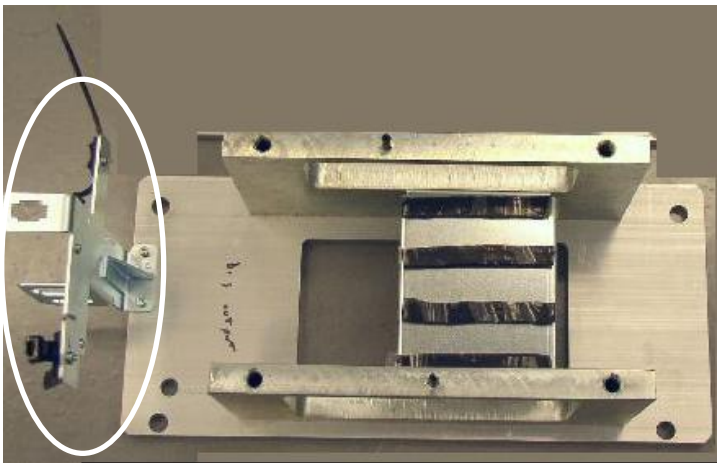


Illustration14: Bulkhead Bracket Installed (P/N 2332018-4)

4. Place the digital power supply mount on the rotating base. The guide pin will only allow the mount to be installed the correct way.

NOTE: Make sure to use the longer M12x40. This is the longer of the two bolts in the kit.

5. Apply loctite 271 and install all 4 M12x40 bolts and torque them to the specification in Table 2.

Table: 2 Torque Specifications			
66.4 N-m	49 lb-ft	588 lb-in	677 kg-cm

6. Proceed to 4.3 Astec Digital Power Supply Installation.

4.4 – ASTEC DIGITAL POWER SUPPLY INSTALLATION

1. Install the Astec digital power supply, using the guide pins to drop it in place.
2. DO NOT use Loctite. Install flat and lock washers and tighten the 4 M12x30 bolts according to the Specifications shown in table 2, using the 10mm Hex wrench. See Illustration 15.



Illustration 15: Digital Power Supply

3. Re-connect J1, J2 on the digital power supply. Use Torque screwdriver to torque all cable jackscrews to the specifications in Table A in Section 1.
4. Rotate the gantry slightly to get better access to the DAS Fan/Htr/DMB connector bracket.
5. Mount cable connectors to Detector Heater Bracket.
6. Follow the sequence shown in Illustration 16 to complete cable connections and proper cable routing.

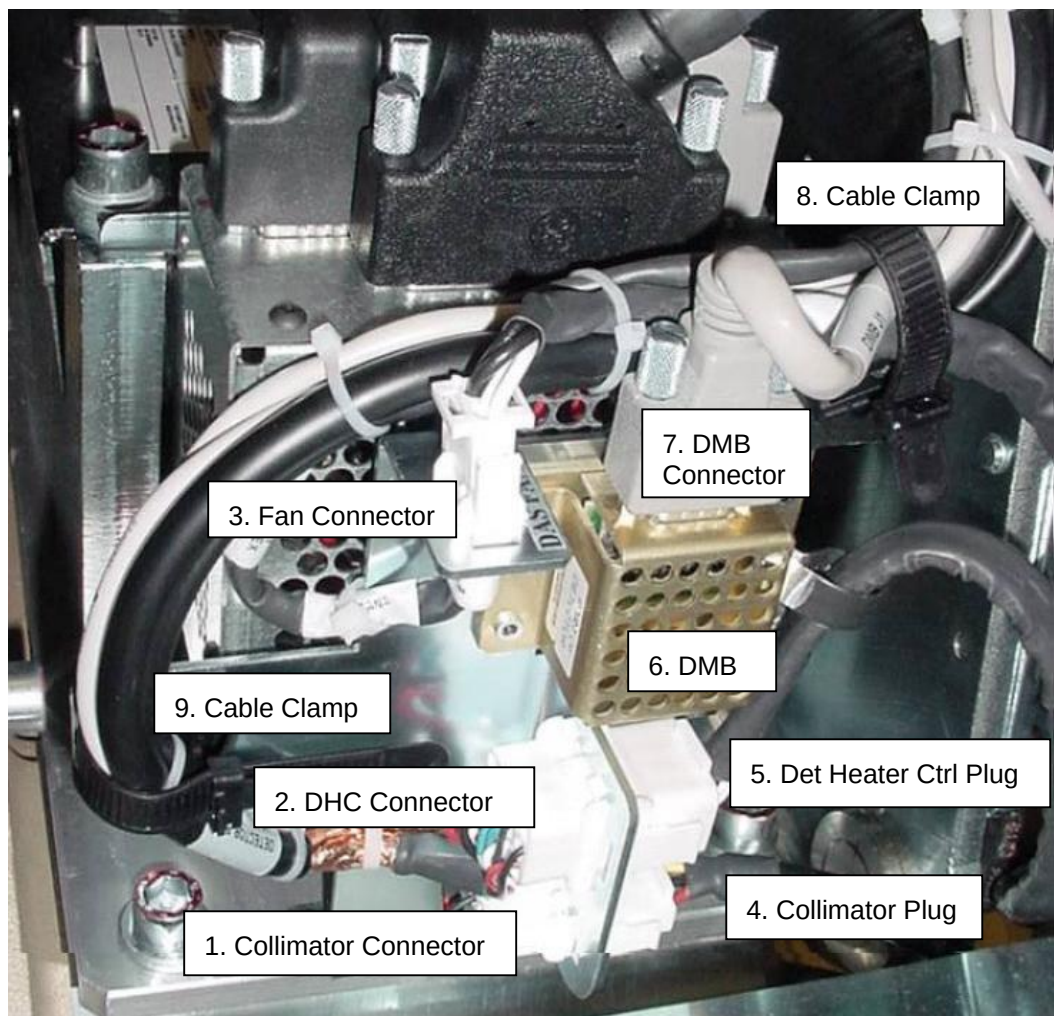


Illustration 16: Astec Digital Power Supply Cable Routing

Note: Use Torque screwdriver to torque all backplane and Power supply cable jackscrews to the specifications in Table A in Section 1.

7. Connect J30, J31, J32, J104 and J33 to the GDAS backplane.
8. Connect J1, J2 and J3 on the Detector Heater Control Board. Route these cables and secure them with the re-usable cable clamps.
9. Reattach the backplane connector restraint bracket for the Right DAS backplane. If necessary, refer back to illustration 7.
10. Adjust the cable harness to allow for gantry rotation. Temporarily secure any loose cables and connectors with tie wraps.
11. Proceed to section 5 GEMS-IT Analog Power Supply Removal.

SECTION 5 – GEMS-IT ANALOG POWER SUPPLY REMOVAL

In this section the GEMS-IT power Supply is removed and the new power supply mount and Astec Analog power supply is installed.

5.1 –GEMS-IT ANALOG POWER SUPPLY REMOVAL

1. Unplug the 24VDC cable, J1 from the ORP.
2. Remove the ORP mount bolts. Let the ORP hang in place. (No need to remove the ORP completely).

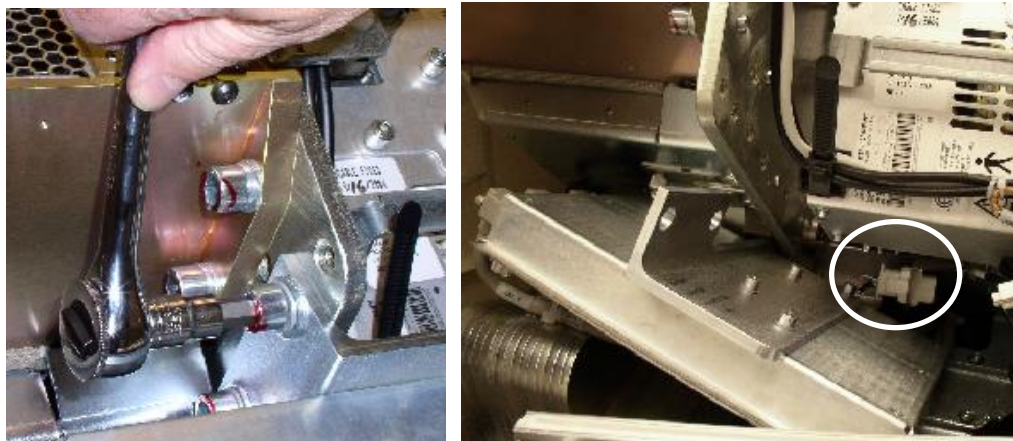


Illustration 17.Remove ORP Mount Bolts To get to Mate-N-Lok Connector

3. Disconnect the plug, (Cable is marked Detector Left, J5) on the Mate-N-Lok connector shown circled, in illustration 17.
4. Cut the tie wrap to free the cable from the bracket. See illustration 18. (This tie wrap will not be reinstalled).



Illustration 18: Cut Tie wrap Holding Cable to Bracket

5. Remove the remaining side of the Mate-N-Lok connector from the bracket completely, by using a small screwdriver to push down on the tabs.
6. Release the Index Lock.

7. Manually rotate the Analog Power Supply to the Left side of the gantry, (Analog Supply will be removed from the Left side of the gantry, over the TGP).
8. Re-apply the Index lock.
9. Remove connectors J1, J2 and J3 from the GEMS-IT Analog power supply.
10. Remove J110 from the GDAS Backplane.
11. Loosen the cable harness brackets from the DAS Backplane mount plate. Cut the tie wraps on the cable harness brackets and remove them completely from the harness. (These will be reused later). See Illustration 19.



Illustration 19: Cable Harness Bracket

12. Loosen the black re-usable cable clamps.
13. Cut the necessary tie wraps to completely remove the short 5VDC Analog to GDAS power cable from the gantry.
14. Disconnect J96 from the GDAS backplane.
15. Disconnect the 5VDC Power Cable:
 - a. For 16 slices systems, this cable will be replaced with a new short cable, (P/N 5124434) from the upgrade kit later. (Note: The existing 5VDC cable will be discarded. (P/N 2363837).
 - b. For 4 and 8 slice systems the existing cable will be reused. (Note: The cable in the kit (P/N 5124434) will NOT be used).

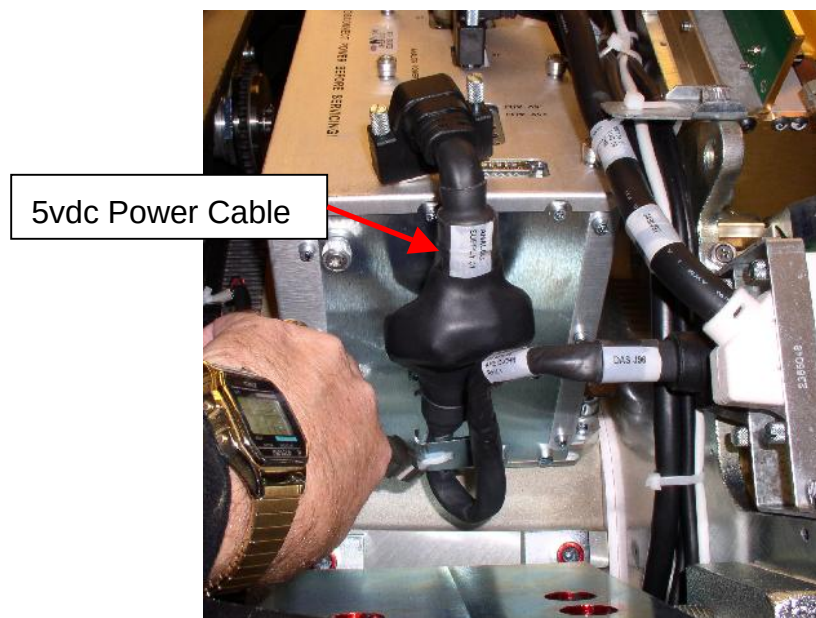


Illustration 20: 5VDC Power cable

16. Remove the top cover of the power supply, using a T25 Torx wrench or medium flat blade screwdriver.
17. Loosen the 3 captive screws holding the fan tray assembly (1 on top and 2 on the side) using a 4mm Hex wrench, T30 Torx wrench or short flat blade screwdriver.
18. Disconnect the Fan tray power plug. If necessary, Refer to Illustration 10 in Section 4.1.
19. Slide the fan tray out of the power supply and set it aside.

20. Using a 10mm Hex bit and necessary extensions completely loosen the 4 mounting bolts. These are captive bolts, so they cannot be lost. If necessary, Refer to Illustration 11 in Section 4.1.
21. Cut the necessary tie wraps to move the DAS Cable harness out of the way for clearance to remove the GEMS-IT Analog Power supply.
22. Remove the GEMS-IT Analog power supply from the gantry.
23. Proceed to Section 6.2 Astec Power Supply Mount Installation.

5.2 –ASTEC ANALOG POWER SUPPLY MOUNT INSTALL

1. Apply Loctite 242 and a 3mm Hex wrench on the two M4x10 (Small screws) provided to install the Detector Heater bracket to the power supply mount. Torque the screws to 2.3 N-m. See illustration 21.

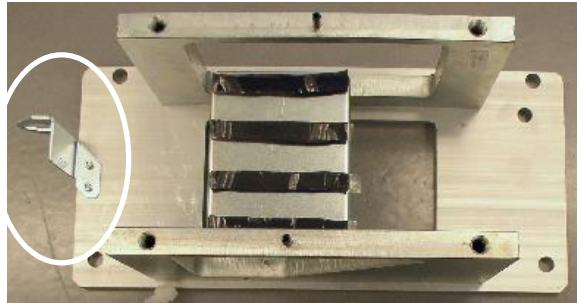


Illustration 20: Detector Heater Bracket Install (P/N 2293442-4)

2. Place the analog power supply mount on the rotating base. Guide pins will only allow for correct installation.
3. Apply Loctite 271. Install all 4 M12x40 bolts. Tighten to torque specifications shown in Table 3.

Table 3: Torque Specifications

66.4 N-m	49 lb-ft	588 lb-in	677 kg-cm
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4. Install the Mate-N-Lok (Cable is marked Detector Left, J5) connector into the new Detector Heater bracket on the analog power supply mount.
5. Plug in the Detector Heater mating plug. (J5).
6. Proceed to 3.3 Astec Analog Power Supply Installation.

5.3 –ASTEC ANALOG POWER SUPPLY INSTALLATION

1. Install the Astec Analog Supply to the new mount in the gantry, the guide pin will only allow one way to install it correctly.
2. DO NOT use Loctite. Install flat and lock washers and tighten the 4 M12x30 bolts using the 10mm Hex wrench according to the specifications in Table 3.
3. Locate the 5V cable in the kit, (P/N 5124434).

Note: Use Torque screwdriver to torque all backplane and Power supply cable jackscrews to the specifications in Table A in Section 1.

- a. Connect J1 to the Astec analog Power supply.
 - b. Connect J96 to the Backplane
4. Attach the restraint bracket found in the kit (P/N 5130042) over connector J1 on the analog power supply.

NOTE:

- 16 slice systems require replacement of the existing 5VDC cable. This cable is part of the kit contents, (P/N 5124434).
 - 4 and 8 Slice systems WILL NOT require replacement of the 5VDC cable, (P/N 2363838). Re-install the existing cable previously removed, and DISCARD the cable (P/N 5124434) that came with the kit.
5. Install the 5VDC Power Cable:
 - a. For 16 slices systems, install the new short cable, (P/N 5124434) from the upgrade kit (Note: Discard the existing 5VDC cable, (P/N 2363838).
 - b. For 4 and 8 slice systems re-install the existing cable, (P/N 2363838). (Note: DO NOT USE the 5VDC cable in the kit (P/N 5124434).
 6. Secure any loose cables temporarily with tie wraps.
 7. Release the Index Lock.
 8. Rotate the gantry to position the Analog supply back to the Right side of the gantry.
 9. Re-apply the Index Lock.
 10. Reattach the ORP and torque the bolts to the specification in table 3. If necessary, refer back to illustration 17.
 11. Do not connect the cable to J1 at this time.
 12. Proceed to Section 6, GDAS Cable Harness Modification.

SECTION 6 – GDAS HARNESS MODIFICATION

6.1 GDAS HARNESS MODIFICATION OVERVIEW

The following illustrations show the extent of the modification in detail refer to these illustrations.

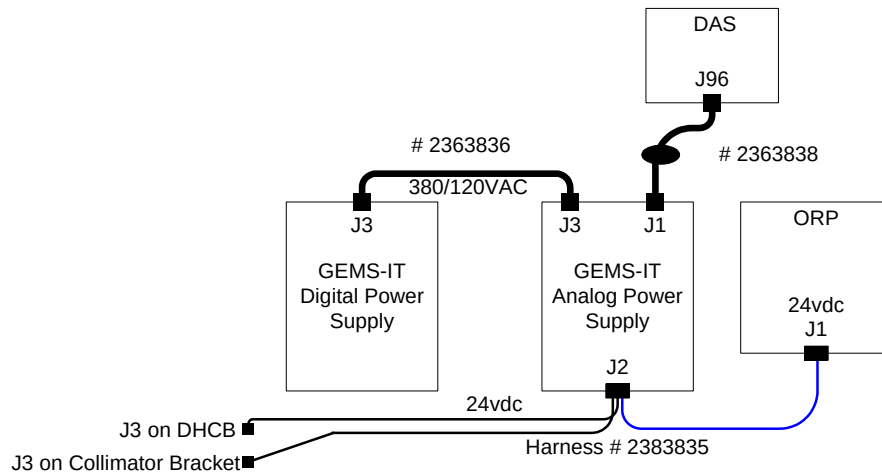
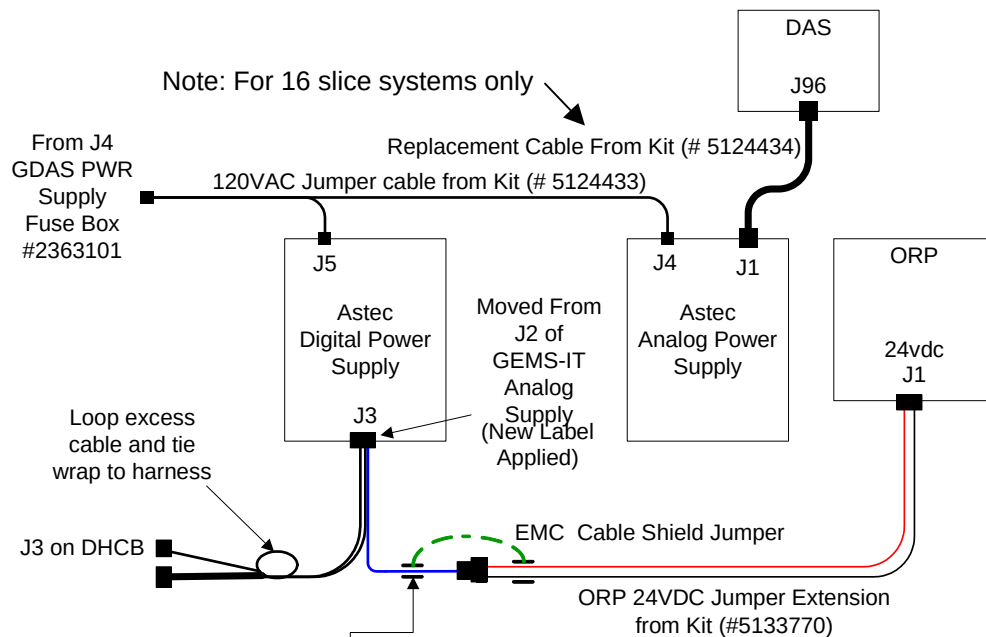


Illustration 21: Affected cables BEFORE DAS Harness Modification



1. Remove tie wrap holding conductive tape to Jumper Extension cable (#5133770).
2. Peel off white backing.
3. align bare wire parallel with exposed shield on ORP cable.
4. Wrap connection with conductive tape.
5. Secure with tie wrap.
6. Install strain relief tie wrap.

Illustration 22: Affected cables AFTER DAS Harness Modification is complete

Process Overview:

- Remove the 380/120VAC cable and discard. (P/N 2363836).
- Re-route the 24VDC supply cable the Digital Power supply side of the gantry.
- Re-label connector J2 on the existing 24VDC harness with new label "24V Digital PS J3".
- Plug newly labeled J3 connector into J3 of Astec Digital Power supply.
- Add jumper/Extension cable, (P/N 513770) to existing harness (P/N 2383835), extending the length of the cable that supplies 24VDC to the ORP.
- Re-route the 24VDC ORP jumper/extension cable.
- Install EMC Shield Jumper from the new 24VDC ORP jumper/Extension cable (P/N 513770) to the existing ORP cable conductive shield.
- Complete all cable connections to GDAS Backplane and new Astec Power Supplies.
- Tie wrap all loose cables to the existing DAS cable harness and reattach all cable Harness restraint brackets.

6.2 GDAS HARNESS MODIFICATION

1. From the Right side of the gantry:
 - a. Disconnect J49 on the Right GDAS Backplane.
 - b. Disconnect J99 on the Center GDAS Backplane.
2. Cut the tie wraps to completely free this cable from the harness. This cable will be reused. (This cable is marked 2287134-4 or 2287134-2).
3. Release the Index Lock.
4. Rotate the gantry if necessary and loosen the black re-usable cable clamp near the 180 degree bracket to free the cables.
5. Re-engage the Index Lock.
6. Disconnect J104 from the Center DAS.
7. Cut all the tie wraps that will free the GEMS-IT digital power supply to GEMS-IT analog power supply 380VAC cable from the cable harness. Remove this cable completely from the harness, (P/N 2363836). Discard this cable. See illustration 23.



Illustration 23: Removal of the 380VAC Power Cable

8. Disconnect J1 on the ORP (24VDC supply).
9. Cut the necessary tie wraps on the DAS harness and extend J2 as far as possible on the 24VDC cable, (P/N 2383835).
10. Push the J2 connector back through to the Left side of the gantry. (Towards digital power supply).

11. From the left side of the gantry, remove the existing label on the 24VDC cable marked "Analog Supply 24VDC "J2" and apply the new label marked "Digital Supply 24VDC J3" from the upgrade kit.

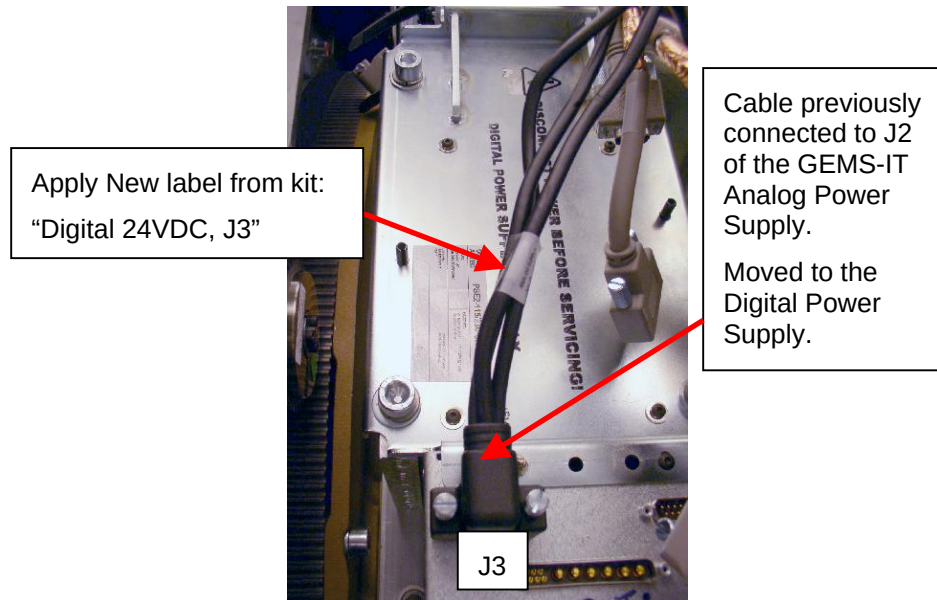


Illustration 24: Re-label and connect cable to Astec Digital Supply J3

12. Plug this cable into J3 of the Digital Supply. Refer to illustration 24.
13. Locate the 24VDC Jumper/Extender cable that came with the kit. (P/N 5133770).

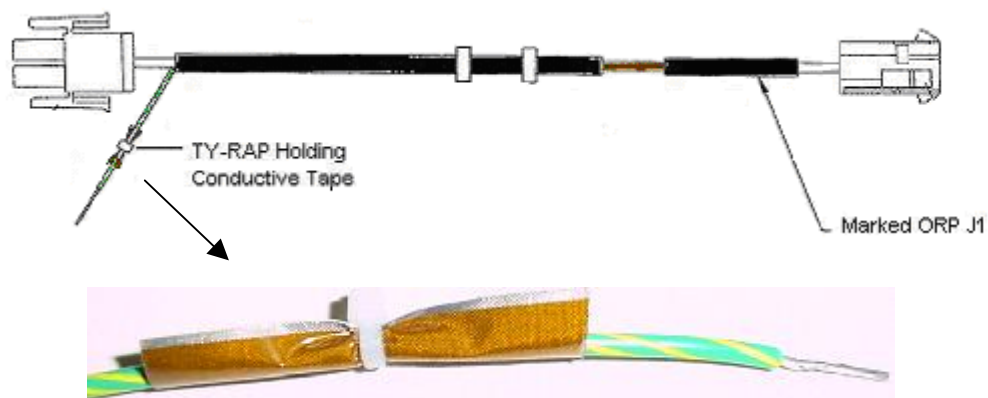


Illustration 25: 24VDC ORP Jumper/Extender Cable (P/N 5133770)

14. Cut the tie wrap holding the conductive tape to the EMC jumper wire found at one end of this cable. Set the conductive tape aside. The conductive tape will be used later, DO NOT LOSE IT.
15. Attach the ORP jumper/extender cable (P/N 5133770) to the original ORP J1 connector on this 24VDC cable. If necessary, refer back to Illustration 22.
16. Feed this jumper/extension cable back to the new analog power supply (RIGHT side of the gantry) following the existing DAS harness. This jumper/extension allows the existing 24VDC cable to be use to supply 24VDC to the ORP. See illustration 26.

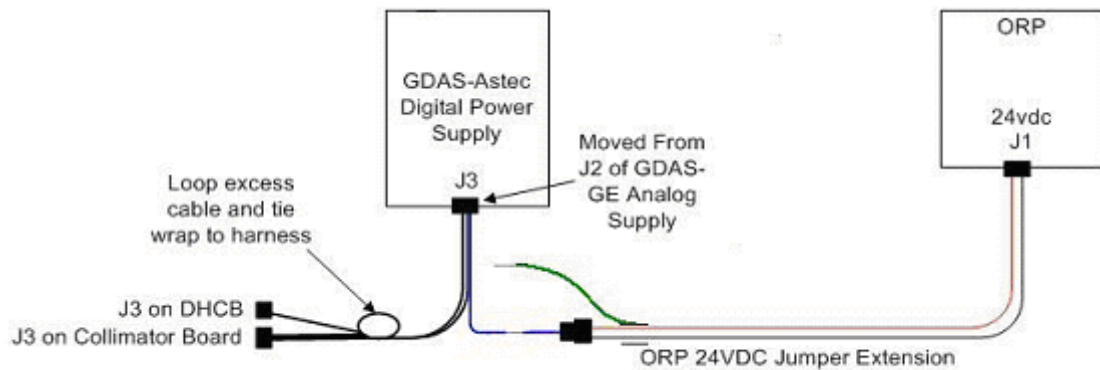


Illustration 26: 24VDC Jumper/Extension cable connection routed to Analog Supply

17. On the Right side of the gantry, arrange the extension cable along the existing DAS cable harness.
18. Plug this extension cable into J1 of the ORP. It will eventually be tie wrapped to the DAS cable harness.
19. Peel off white backing from conductive tape removed in step 14.
20. Align bare wire parallel to cable at the exposed shield of the original ORP cable. (No need to wrap the wire around the shield. Contact is all that is required).



Illustration 27: Example of Shield Jumper/EMC Shield Contact

21. Wrap the connection with conductive tape.
22. Secure connection with tie wrap.
23. Add a second tie wrap as a strain relief. Refer to Illustration 28.

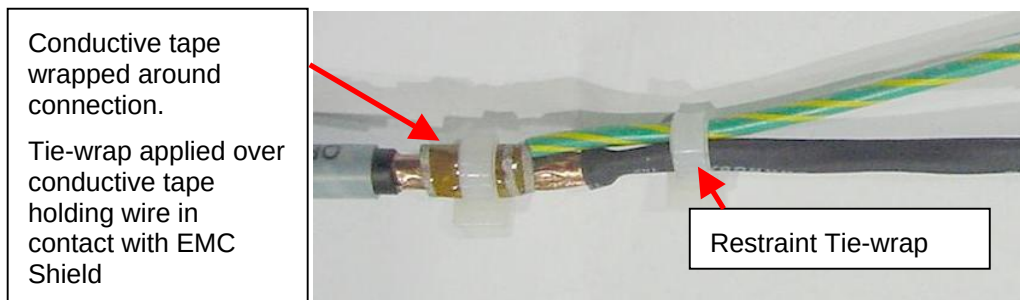


Illustration 28: EMC Jumper connected and restrained

24. From the Right side of the gantry reinstall the cable marked 2287134-4 or 2287134-2.

Note: Use Torque screwdriver to torque all backplane and Power supply cable jackscrews to the specifications in Table A in Section 1.

- a. Connect J49 on the Right GDAS Backplane.
- b. Connect J99 on the Center GDAS Backplane.
25. Connect J104 to the Center Backplane.
26. Locate the new 120VAC Cable in the kit (P/N 5124433). It has three connectors.
27. Connect one connector as marked, to J4 of the DAS Fuse Box.

28. Connect the plug as marked, to J5 of the Digital Power Supply
29. Route the other end of the 120VAC cable along the existing DAS harness back over to the RIGHT side of the gantry and plug it into J4 of the Astec Analog Power Supply.
30. Route the cables to the DHCB and AC Cable from the fuse box, and secure them with the re-usable cable clamps mounted on the new digital power supply mount.
31. Reinstall the cable Harness brackets on the right side of the gantry.



Illustration 29: Cable Harness Bracket

32. Connect the remaining ORP cables and any backplane cables removed on the RIGHT and Center DAS Backplane.
33. Connect GDAS cable J110 to the Right GDAS Backplane.
34. Route the cables through the re-usable Cable clamps and secure them.
35. Dress all cables along the existing DAS harness and tie wrap them to the main DAS harness.
36. Secure the DAS harness to the cable harness brackets. Refer to illustration 32.
37. Return to the Left side of the gantry.
38. Reconnect any remaining cables removed.
39. Reinstall the two cable Harness brackets to the DAS backplane mounting plate.
40. Tie wrap the cables back into the main DAS harness dressing the cables as you go. Loop any extra length in the existing 24VDC cable and tie wrap it to the main harness.
41. Tie wrap the complete DAS Harness to the Cable Harness brackets as shown in Illustration 30.

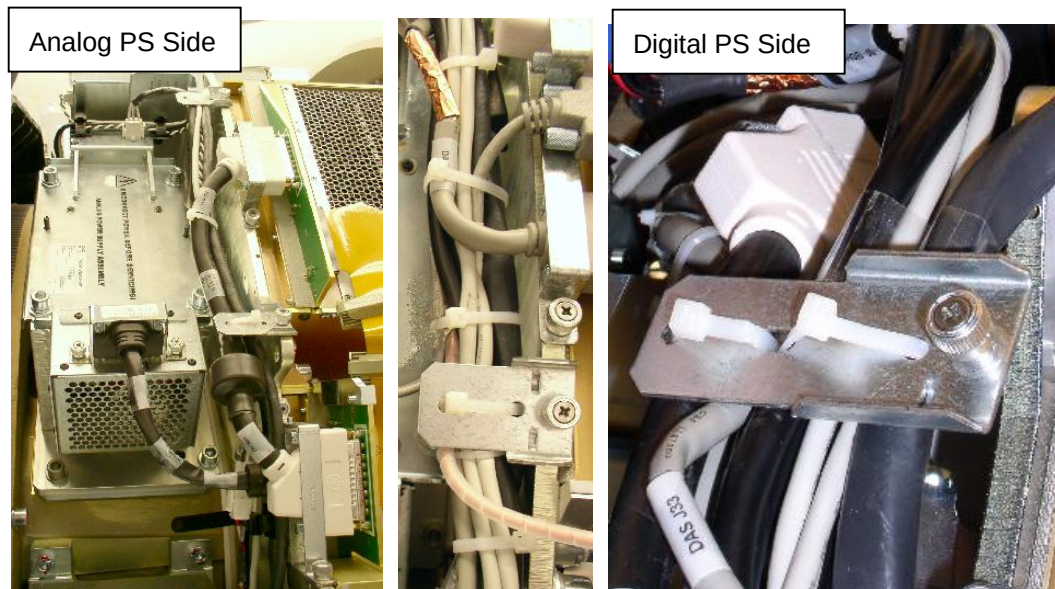


Illustration 30: Cable routing and Tie wrapped Cable Harness Brackets

42. Continue to add tie wraps at intervals, bundling the DAS harness back into its original position.
43. Finish restraining the cables with using the re-usable black cable connectors.
44. Re-install the EMC Cover on the Left side of the DAS chassis.
45. MANUALLY rotate the gantry several times while inspecting the cables removed, to make sure that they are properly secured to the rotating side of the gantry, and are not loose or interfering with rotation.
46. Proceed to section 7- Finalization.

SECTION 7 – FINILIZATION

Note: Gantry balance has not been affected by this procedure. The replacement parts duplicate the same mass as the older GEMS-IT Power Supply configuration.

1. Verify the cable connections are complete using table 4.

Table 4: Cable connection Checklist

√	Location	Connector
	Analog Supply	J1, J4
	ORP	J1, J5
	Right DAS Chassis	J110
	Center DAS Chassis	J96, J97, J99, J104
	Digital Supply	J1, J2, J3, J5
	DMB	J1
	DHCB	J1, J2, J3
	Left DAS Chassis	J30, J31, J32, J33, J49

2. Remove the Index Lock and all safety equipment.
3. From the Right side of the gantry, turn on the Axial Drive, HVDC, and 120VAC Switches.
4. Replace the top covers and connect the top cover fans.
5. Adjust 5V supplies output. Refer to [GDAS Power supply Adjustment](#) procedure.
6. Replace side covers.
7. Perform a Fastcal to ensure everything is functioning normally.
8. Perform a basic scan to ensure the system is functioning normally.



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